

Track A Documentation

Chapter 3: Apoptosis — ODE Model & Book Reproduction

Schiesser, W.E. (2014) — Spring 2026

1. Biological Background

Apoptosis is the process of **programmed cell death** — the body's built-in mechanism for eliminating damaged, aged, or unwanted cells in a controlled manner. Unlike necrosis (uncontrolled cell death caused by injury), apoptosis is an ordered, energy-dependent process that plays a fundamental role in organism development, immune function, and tissue homeostasis.

The process is mediated by a cascade of molecular signals. When a cell receives a death signal — such as **hypoxia (low oxygen), radiation, heat, or cytotoxic drugs** — it activates specific gene products that execute cell death. The key players are:

- **HIF-1 (Hypoxia Inducible Factor):** a transcription factor activated under low-oxygen conditions. It drives angiogenesis (new blood vessel formation) and cellular stress responses.
- **O₂ (Oxygen):** oxygen level directly regulates HIF activity — when oxygen is high, HIF is suppressed. Low oxygen activates HIF and drives the hypoxia response.
- **p300 Co-activator:** a transcriptional co-activator that interacts with both HIF-1 and p53 to regulate gene expression. It modulates the balance between cell survival and death.
- **p53 (Tumor Suppressor Gene):** known as the "guardian of the genome". When DNA is damaged or cellular stress is detected, p53 is activated and promotes either DNA repair or apoptosis. Mutations in p53 are found in over 50% of human cancers.
- **Caspase (CASP):** the executioner proteins that carry out apoptosis. Once caspases are activated (primarily by p53 signaling), they dismantle the cell from within. Higher caspase levels = more apoptotic activity.
- **K⁺ (Potassium Ions):** potassium efflux (outflow) from the cell is one of the earliest events in apoptosis. Loss of K⁺ triggers cell shrinkage and promotes caspase activation.

In cancer, this process goes wrong — cancer cells **resist apoptosis**, allowing them to survive and proliferate indefinitely. Mathematical modeling of apoptosis is therefore critical for understanding cancer biology and developing therapeutic strategies.

2. The ODE Model

This model is based on the paper by **Laise et al.**, as presented in Schiesser Chapter 3. It describes apoptosis regulation through **six coupled nonlinear ordinary differential equations**, one for each molecular species. The model captures the interactions between HIF-1, oxygen, p300, p53, caspase, and potassium ions as they evolve toward a biological equilibrium.

2.1 State Variables

The six dependent variables of the model are listed in Table 1. All variables represent molecular concentrations and must remain non-negative at all times.

Variable	Symbol	Unit	Biological Meaning
y[0]	y_hif	Normalised conc.	Hypoxia Inducible Factor (HIF-1) — activated by low oxygen, drives hypoxia response and angiogenesis
y[1]	y_o2	Normalised conc.	Oxygen concentration — suppresses HIF when high; low O ₂ triggers apoptotic cascade
y[2]	y_p300	Normalised conc.	p300 Co-activator — regulates gene expression by interacting with HIF and p53
y[3]	y_p53	Normalised conc.	p53 Tumor Suppressor Gene — "guardian of the genome"; activates caspase when damage detected
y[4]	y_casp	Normalised conc.	Caspase — the executioner protein; higher concentration = more apoptotic activity
y[5]	y_kp	Normalised conc.	Potassium Ions (K ⁺) — early apoptotic signal; efflux triggers cell shrinkage and caspase activation

Table 1. State variables of the apoptosis ODE model (Schiesser Table 3.1, p.147).

2.2 Initial Conditions

The initial conditions (ICs) are given in Schiesser Table 3.2 (p.147). Five of the six variables start at zero; only **y_hif starts at 1**. These ICs are artificial — they do not represent a specific physical state. They were chosen because the study focuses on the **final equilibrium** rather than the transient behaviour, and the equilibrium is IC-independent (as demonstrated in Section 4.2).

Variable	Value at t = 0	Physical Justification
y_hif(0)	1	Non-zero start — system begins under hypoxic/stressed condition
y_o2(0)	0	Oxygen starts at zero — complete hypoxic state initially
y_p300(0)	0	No co-activator initially
y_p53(0)	0	No tumor suppressor activity initially
y_casp(0)	0	No caspase (no apoptosis) initially
y_kp(0)	0	No potassium efflux initially

Table 2. Initial condition vector (Schiesser Table 3.2, p.147).

2.3 Time Span

The simulation runs from **t = 0 to t = 100** in dimensionless time units. The time units are not specified in the original model — they are dimensionless because none of the parameters carry units of time. A total of **101 output points**

are used (every 1 unit), matching the book's `seq(from=0, to=100, by=1)` specification. The solution reaches equilibrium well before $t = 100$.

3. Model Parameters

The parameter set is taken directly from **Schiesser Table 3.3 (p.148)**, which in turn is from Laise et al. All 14 parameters are non-negative. Their roles are illustrated in the interaction diagram (Fig. 3.1 of the book).

Symbol	Value	Appears in	Biological Role
a_hif	1.52	Eq. 3.1	Constant production rate of HIF — drives the system away from zero ICs; ensures HIF is always being produced
a_o2	1.80	Eq. 3.2	Constant production rate of O ₂ — oxygen is continuously supplied to the cell
a_p53	0.05	Eq. 3.4	Constant basal production rate of p53 — small but nonzero, ensuring p53 never fully disappears
a_3	0.90	Eqs. 3.1, 3.2	Rate of HIF-O ₂ interaction — when both HIF and O ₂ are present, they react and reduce each other
a_4	0.20	Eqs. 3.1, 3.2, 3.3	Rate of HIF-p300 interaction — HIF binds to p300; this depletes HIF and p300 but produces O ₂
a_5	0.001	Eqs. 3.3, 3.4	Rate of p300-p53 interaction — very small; p300 and p53 interact to deplete each other slowly
a_7	0.70	Eq. 3.1	Rate of p53-HIF interaction — p53 promotes HIF degradation (negative feedback)
a_8	0.06	Eq. 3.3	Constant production rate of p300 — small steady source ensuring p300 is always present
a_9	0.10	Eqs. 3.4, 3.5	p53 natural decay rate AND p53-to-caspase conversion rate — p53 activates caspase before decaying
a_10	0.70	Eq. 3.6	Rate of caspase-K ⁺ interaction — active caspase consumes K ⁺
a_11	0.20	Eqs. 3.2, 3.6	O ₂ natural decay rate AND O ₂ -to-K ⁺ production rate —

			oxygen drives K ⁺ production
a_12	0.10	Eq. 3.5	Constant caspase source — basal caspase activity independent of p53
a_13	0.10	Eq. 3.5	Caspase natural decay rate — caspase is continuously degraded
a_14	0.05	Eq. 3.6	K ⁺ natural decay rate — potassium ions leak back into the cell over time

Table 3. Parameter set for the apoptosis ODE model (Schiesser Table 3.3, p.148).

4. The Six ODEs — Equations and Term-by-Term Explanation

The six equations below are taken directly from Schiesser eqs. (3.1)–(3.6), p.146–147. Each equation governs one concentration variable. Note that **all equations are smooth and nonlinear** — unlike the diabetes chapter, there are no piecewise switches here. The nonlinearity arises from the product terms (e.g., $y_{hif} \times y_{o2}$) which represent biochemical interactions requiring both molecules to be present simultaneously.

Derivative	Equation	Variable Governed
dy_{hif}/dt	$= a_{\text{hif}} - a_3 y_{\text{o2}} y_{\text{hif}} - a_4 y_{\text{hif}} y_{\text{p300}} - a_7 y_{\text{p53}} y_{\text{hif}}$	HIF concentration
dy_{o2}/dt	$= a_{\text{o2}} - a_3 y_{\text{o2}} y_{\text{hif}} + a_4 y_{\text{hif}} y_{\text{p300}} - a_{11} y_{\text{o2}}$	Oxygen concentration
dy_{p300}/dt	$= a_8 - a_4 y_{\text{hif}} y_{\text{p300}} - a_5 y_{\text{p300}} y_{\text{p53}}$	p300 co-activator
dy_{p53}/dt	$= a_{\text{p53}} - a_5 y_{\text{p300}} y_{\text{p53}} - a_9 y_{\text{p53}}$	p53 tumor suppressor
dy_{casp}/dt	$= a_{12} + a_9 y_{\text{p53}} - a_{13} y_{\text{casp}}$	Caspase concentration
dy_{kp}/dt	$= -a_{10} y_{\text{casp}} y_{\text{kp}} + a_{11} y_{\text{o2}} - a_{14} y_{\text{kp}}$	Potassium ions

Table 4. The six ODEs of the apoptosis model (Schiesser eqs. 3.1–3.6).

4.1 Term-by-Term Explanation

Equation 3.1 — HIF (dy_{hif}/dt)

- $+a_{\text{hif}}$: constant source (1.52) — HIF is continuously produced
- $-a_3 * y_{\text{o2}} * y_{\text{hif}}$: HIF- O_2 interaction — when oxygen is high, HIF is degraded rapidly (negative feedback from oxygenation)
- $-a_4 * y_{\text{hif}} * y_{\text{p300}}$: HIF-p300 interaction — HIF binds p300, depleting both
- $-a_7 * y_{\text{p53}} * y_{\text{hif}}$: p53 suppresses HIF — when p53 rises (indicating damage), it promotes HIF degradation

Equation 3.2 — Oxygen (dy_{o2}/dt)

- $+a_{\text{o2}}$: constant source (1.80) — oxygen is continuously supplied
- $-a_3 * y_{\text{o2}} * y_{\text{hif}}$: same HIF- O_2 interaction as eq.3.1 — oxygen consumed when reacting with HIF
- $+a_4 * y_{\text{hif}} * y_{\text{p300}}$: HIF-p300 interaction *produces* oxygen (opposite sign from eq.3.1) — when HIF binds p300, O_2 is released
- $-a_{11} * y_{\text{o2}}$: linear O_2 decay — oxygen is consumed by cell metabolism

Equation 3.3 — p300 (dy_{p300}/dt)

- $+a_8$: small constant source (0.06) — p300 is slowly but continuously produced
- $-a_4 * y_{hif} * y_{p300}$: HIF-p300 interaction depletes p300
- $-a_5 * y_{p300} * y_{p53}$: p300-p53 interaction — very slow ($a_5=0.001$), depletes p300

Equation 3.4 — p53 (dy_{p53}/dt)

- $+a_{p53}$: small constant source (0.05) — basal p53 production
- $-a_5 * y_{p300} * y_{p53}$: p300-p53 interaction depletes p53
- $-a_9 * y_{p53}$: linear p53 decay — p53 is naturally degraded AND converts to caspase signal

Equation 3.5 — Caspase (dy_{casp}/dt)

- $+a_{12}$: constant basal caspase source (0.10) — some caspase is always present
- $+a_9 * y_{p53}$: p53 activates caspase — the more p53, the more caspase is produced (apoptosis signal)
- $-a_{13} * y_{casp}$: caspase natural decay (inhibited by anti-apoptotic proteins in real cells)

Equation 3.6 — Potassium Ions (dy_{kp}/dt)

- $-a_{10} * y_{casp} * y_{kp}$: caspase consumes K^+ — as apoptosis proceeds, caspase drives K^+ efflux
- $+a_{11} * y_{o2}$: oxygen drives K^+ production — oxygenation promotes ion transport
- $-a_{14} * y_{kp}$: linear K^+ decay — potassium ions naturally return to baseline

5. Simulation Cases

The book runs the model under different configurations to study its behaviour. Two cases are essential for Track A.

Case	y_hif(0)	a_12	Purpose
1 ★	1 (non-zero)	0.10 (constant)	Base case — standard model run. This is the ground truth used by all other tracks for comparison.
2	0 (zero)	0.10 (constant)	IC variation — proves the equilibrium is independent of initial conditions. The model reaches the same steady state regardless of where it starts.
3	1 (non-zero)	$a_{12} * e^{-0.1t}$ (decaying)	ODE variation — models cancer behaviour where the caspase source fades over time. Caspase drops significantly, K^+ rises. Not required for Track A but included for completeness.

Table 5. Simulation cases. ★ = primary case / ground truth.

6. Numerical Results — Reproduction Validation

6.1 Solution at Selected Time Points (Case 1)

Table 6 shows the Python LSODA solution at selected time points, matching the format of **Schiesser Table 3.4 (p.164)**. The system evolves from the initial conditions toward equilibrium.

t	y_hif	y_o2	y_p300	y_p53	y_casp	y_kp
0.0	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5.0	1.1827	1.3592	0.1625	0.1967	0.4386	0.5995
10.0	0.7753	1.9395	0.2636	0.3158	0.7642	0.6736
25.0	0.5420	2.6645	0.4706	0.4576	1.2735	0.5706
50.0	0.5038	2.8407	0.5771	0.4940	1.4708	0.5267
75.0	0.4996	2.8618	0.5950	0.4968	1.4943	0.5223
100.0	0.4991	2.8646	0.5978	0.4970	1.4968	0.5219

Table 6. Python LSODA numerical output for Case 1 (base case). Matches Schiesser Table 3.4.

6.2 Equilibrium Validation

Table 7 compares the Python computed equilibrium at $t = 100$ against the book values. All absolute errors are below 4×10^{-5} , confirming successful reproduction.

Variable	Python (t=100)	Book Table 3.4	Absolute Error	Status
y_hif	0.499073	0.4991	0.000022	✓ Match
y_o2	2.864629	2.8646	0.000029	✓ Match
y_p300	0.597827	0.5978	0.000027	✓ Match
y_p53	0.496986	0.4970	0.000014	✓ Match
y_casp	1.496836	1.4968	0.000036	✓ Match
y_kp	0.521918	0.5219	0.000018	✓ Match

Table 7. Validation of Python reproduction against Schiesser Table 3.4. All errors < 4×10⁻⁵.

6.3 IC Independence Verification (Cases 1 vs 2)

Table 8 compares the equilibrium values at t = 100 for Case 1 (y_hif(0)=1) and Case 2 (y_hif(0)=0). Both converge to **identical values**, confirming that the model reaches a unique stable equilibrium independent of the chosen initial conditions. This is an important stability property of the nonlinear ODE system.

Variable	Case 1 (y_hif ₀ =1)	Case 2 (y_hif ₀ =0)	Difference
y_hif	0.4991	0.4991	< 1×10 ⁻⁴
y_o2	2.8646	2.8646	< 1×10 ⁻⁴
y_p300	0.5978	0.5978	< 1×10 ⁻⁴
y_p53	0.4970	0.4970	< 1×10 ⁻⁴
y_casp	1.4968	1.4968	< 1×10 ⁻⁴
y_kp	0.5219	0.5219	< 1×10 ⁻⁴

Table 8. IC independence — both cases reach the same equilibrium (Schiesser Section 3.6).

8. Ground Truth — Values for Other Tracks

The LSODA solution from Case 1 is the ground truth that **all other tracks compare against**. The key values to use for RMSE computation are:

Variable	Equilibrium (t=100)	Peak Value	Notes for comparison
y_hif	0.4991	~1.68 at t~3	Starts high, drops sharply — test early accuracy
y_o2	2.8646	2.8646 (monotone)	Monotonically increasing — easy to track
y_p300	0.5978	0.5978 (monotone)	Slowly increasing — smooth curve
y_p53	0.4970	0.4970 (monotone)	Slowly increasing — smooth curve
y_casp	1.4968	1.4968 (monotone)	Monotonically increasing — smooth curve
y_kp	0.5219	~0.67 at t~5	Slight overshoot then decay — test peak accuracy